

SP-4 Phase F — Clone-with-new-name actually searchable (detailed design)

Spec date: 2026-05-13 **Parent SP:** docs/superpowers/specs/2026-05-12-sp4-multi-provider-redesign.md **Prerequisite:** Phases A + B + C + D complete (HEAD 657f74c1). **Status:** detailed-design locked. Implementation plan in docs/superpowers/plans/2026-05-13-sp4-phase-f-implementation.md.

Goal (from parent SP-4 design)

Phase F — Provider clone-with-new-name. UI clone affordance. Domain support for "multiple instances of the same provider type with different mirror sets". Challenge for clone+search the clone.

Current state audit (post Phase D, HEAD 657f74c1)

Surface	What's there
Phase B CloneSection UI	ProviderConfigAction.ConfirmClone(display primaryUrl) → CloneProviderUseCase → ClonedProviderDao row + SyncOutbox enqueue
Phase A ClonedProviderDao / ClonedProviderEntity	Persists (syntheticId, sourceTrackerId, displayName, primaryUrl) Returns base + synthetic ClonedTrackerDescri (capabilities + AuthType + apiSupported all del to source; trackerId + displayName + baseUrls a clone's)
LavaTrackerSdk.listAvailableTrackers()	
LavaTrackerSdk.search / multiSearch / streamMultiSearch	All call registry.get(id, MapPluginConfig()) synchronously

Surface	What's there
Source tracker clients (e.g. RuTrackerClient)	Construct their HTTP/Ktor stack with <code>descriptor.baseUrls</code> baked in at <code>factory.create</code>

What this is and isn't

This phase is **F.1** — close the SDK crash + tag results with clone id so the multi-provider UI correctly displays "from RuTracker EU" instead of "from RuTracker". Searches against a clone route to the **source** tracker's TrackerClient and use the **source's** baseUrls — the clone's `primaryUrl` is recorded in the DAO but NOT yet routed.

F.2 (owed, separate phase) wires the URL override end-to-end: per-clone MirrorManager OR `PluginConfig.baseUrlOverride` support across factories, so a search against the "RuTracker EU" clone hits `rutracker.eu` instead of `rutracker.org`.

Honest disclosure: when the user creates a clone, the ConfirmClone Toast text changes from "Cloned" to "Cloned (URL routing pending — searches use source URLs until Phase F.2)" so the user is not misled about what the clone is doing today. This is the §6.J anti-bluff mechanic: ship value, name the gap, no false promise.

Design decisions (locked)

F-D1 — Resolution at the SDK seam, not in the registry

The clone's synthetic id is NOT registered as a separate factory. Instead, every SDK call that does `registry.get(id, MapPluginConfig())` goes through a new private suspend helper `clientFor(id: String): TrackerClient` that:

- If `id` is a synthetic clone (looked up in `ClonedProviderDao`): construct a wrapping `ClonedRoutingTrackerClient` that
 - Reports the clone's descriptor (via `ClonedTrackerDescriptor`)
 - Delegates `getFeature(...)` to a **clone-aware shim** that wraps the source feature: calls the source feature's method, then re-tags result items with the clone's synthetic id.
- Else: `registry.get(id, MapPluginConfig())` as today.

Why at the SDK seam, not at the registry: keeping the synthetic-id resolution out of the registry preserves the registry's contract ("if you call register, you can call get"; clones live outside that contract — they're a Lava-domain composition on top).

F-D2 — Result re-tagging

`SearchableTracker.search(...)` returns `SearchResult(items, totalPages, currentPage)` where each `TorrentItem` carries `trackerId`. The clone-shim wraps the source feature and re-maps items to have `trackerId = clone.syntheticId` so the multi-provider UI's per-provider grouping shows the clone's display name correctly.

Same pattern for `BrowsableTracker.browse(...)` → `BrowseResult` items, and `FavoritesTracker.list()` items.

F-D3 — `getActiveClient` also resolved

`LavaTrackerSdk.search / browse / getTopic / login / checkAuth / logout / downloadTorrent / getMagnetLink / addFavorite / removeFavorite / getCommentsPage / addComment / getForumTree / getFavorites` ALL call `getActiveClient()` which does `registry.get(activeTrackerId, ...)`. If a user marks a clone as active (legacy `switchTracker(...)` — now `@Deprecated`), every one of those crashes today.

`getActiveClient()` is refactored to a suspend `getActiveClientSuspend()` that uses `clientFor(activeTrackerId)`. The non-suspend overload is kept as a deprecated fallback that calls `runBlocking { ... }` (consistent with the existing `listAvailableTrackers()` pattern that already does this for `ClonedProviderDao.getAll()`).

F-D4 — UI disclosure of the F.2 gap

`ProviderConfigSideEffect.ShowToast("Cloned")` becomes `ShowToast("Cloned (URL routing pending — searches use source URLs)")`. No new UI surfaces; just honest copy.

F-D5 — `UserMirrorDao` keyed by clone id

When the user adds a custom mirror for a clone (via Phase B's `MirrorsSection`), the `UserMirrorEntity.trackerId` field carries the clone's synthetic id. The infrastructure already supports this — DAO doesn't care if the `trackerId` is a clone or not. F.2 will read this DAO to construct the per-clone `MirrorManager`.

Affected surfaces

Files to add

- `core/tracker/client/src/main/kotlin/lava/tracker/client/ClonedRoutingTrackerClient.kt` — the wrapping client.
- `core/tracker/client/src/test/kotlin/lava/tracker/client/LavaTrackerSdkCloneSearchTest.kt` — unit tests: search a clone returns items tagged with clone id; the source's parser ran (asserted by the items' contents).

Files to modify

- `core/tracker/client/src/main/kotlin/lava/tracker/client/LavaTrackerSdk.kt`:
 - Add `private suspend fun clientFor(id: String): TrackerClient`.
 - Replace every `registry.get(id, MapPluginConfig())` call with `clientFor(id)`. (~12 call sites: `search`, `browse`, `getTopic`, `getTopicPage`, `getCommentsPage`, `addComment`, `downloadTorrent`, `getMagnetLink`, `getForumTree`, `getFavorites`, `addFavorite`, `removeFavorite`, `login`, `checkAuth`, `logout`, `multiSearch`, `streamMultiSearch`).
 - `getActiveClient()` → `getActiveClientSuspend()` for callers that already suspend; the non-suspend version is kept for the internal `SwitchingNetworkApi` compat path.
- `feature/provider_config/src/main/kotlin/lava/provider/config/ProviderConfigViewModel.kt` — Toast copy updated.
- `docs/CONTINUATION.md` — Phase F.1 landed; F.2 owed.

Challenge Tests

C34 (new) `CloneAndSearchClonedProviderTest.kt`:

1. Start on Menu → tap RuTracker.org provider row → ProviderConfig → tap "Clone provider..." → fill name "RuTracker EU" + URL "<https://rutracker.eu>" → Confirm.
2. Toast "Cloned (URL routing pending...)" displayed.
3. Press back → Menu now shows two RuTracker rows: original + "RuTracker EU".
4. Tap "RuTracker EU" → ProviderConfig opens correctly.

Operator-bound (gating emulator).

Falsifiability rehearsal for the unit test:

Unit test 1 asserts that `sdk.multiSearch(clone-id)` does NOT throw (today it does throw `IllegalArgumentException`). Mutation: revert `clientFor(id)` back to `registry.get(id, MapPluginConfig())`. Test fails with the same `IllegalArgumentException`.

Unit test 2 asserts that returned items carry the CLONE's synthetic id, not the SOURCE's id. Mutation: in the clone-shim, return source items unmodified (skip the re-tag). Test fails — the items' `trackerId` is the source's id.

Anti-bluff posture

Per §6.J / §6.L:

1. The Toast copy "(URL routing pending — searches use source URLs)" disclosures the F.2 gap at the moment the user creates the clone — no false promise about the URL override.
2. Unit tests assert on user-observable outcomes: no crash, items tagged with clone id, source parser actually ran.
3. CONTINUATION.md explicitly names F.2 as owed; the SP-4 sequence record includes the gap.
4. F.1 alone IS a complete user-visible improvement: today the search path against clones is a crash; after F.1 the clone appears as a normal-looking provider that returns results. The URL gap is documented, not hidden.

Out of scope for Phase F (this is F.2 owed)

- Per-clone `MirrorManager` wiring so search HTTP traffic actually hits the clone's `primaryUrl`.
- Result-content variation when source URLs are degraded (Phase F.2 lets clones be a failover surface).
- Removing the Toast copy disclosure (it stays until F.2 ships).

Operator's invariants reasserted

- §6.J: searching a clone now produces a real, user-visible outcome (items in the search-result list, correctly tagged with the clone's display name).
- §6.Q: no UI layout changes; `ProviderConfig` screen is unchanged structurally.
- §6.R: no new hardcoded literals.
- §6.S: CONTINUATION updated in implementation commit.
- §6.W: pushed to both mirrors in lockstep.

Implementation plan

See docs/superpowers/plans/2026-05-13-sp4-phase-f-implementation.md — 5 tasks, ~15 steps.